

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

SEPTEMBER EXAMINATION

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS

TUESDAY, 3 OCTOBER 2023

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGY

Instruction to Candidates:

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL THREE (3)** questions in **Section A** and choose **ONLY ONE (1)** question in **Section B**.

Should a candidate answer more than **ONE (1)** question in section B, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are not allowed to use calculator.

Answer questions only in the answer booklet provided.

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Section A [Total: 75 marks]. This section consists of THREE (3) COMPULSORY questions.**

- Q1. (a) The relational schema shown below is part of e-book library management database. The primary keys are highlighted in bold.

Patron (**patID**, patFname, patLname, patType)

Checkout (**checkNum**, bookNo, patID, checkOutDate, checkDueDate, checkInDate)

Book (**bookNo**, bookTitle, bookYear, bookCost, bookSubject, auID)

Writes (**bookNo**, **auID**)

Author (**auID**, auFname, auLname, auBirthYear)

- (i) Write **ONE (1)** Structured Query Language (SQL) statement to display the book title, cost, and year of publication for every book in the system. Arrange the results by book title in ascending order. (3 marks)
- (ii) Write **ONE (1)** SQL statement to show the author ID, first and last name, book title of all books with the subject "Cloud". (4 marks)
- (iii) Write **ONE (1)** SQL statement to display the book number, title, and number of times each book has been checkout. Limit the results to books that have been checkout more than five times. Sort the results in descending order by number of times checked out. (5 marks)
- (b) The Gig Mfg Construct database stores data for a consulting company that tracks all charges to the projects. The charges are based on the hours each employee works on the project.

Table 1.1: Partial records of Job relation

JobCode	JobDescription	JobChg Hour
500	System Analyst	96.75
501	Mechanical Engineer	67.90
502	Clerical Support	26.87
503	Applications Designer	48.10
504	Database Engineer	125.00

Table 1.2: Partial records of Project relation

Proj No	ProjName	ProjValue	Emp No
15	Evergreen	145 500.0	103
18	Amber Wave	35 000.0	104
22	Rolling Tide	80 500.0	102
25	Starflight	260 000.0	101

Table 1.3: Partial records of Employee relation

EmpNo	EmpName	EmpHireDate	JobCode	EmpYears
101	John	08-Nov-10	500	10
102	David	12-Jul-89	504	23
103	June	01-Dec-96	503	13
104	Alice	22-Aug-91	502	16
105	William	18-Jul-97	501	12
106	Maria	22-Jun-14	503	8

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q1. (b)(Continued.)***Table 1.4: Partial records of Assignment relation*

AsgnNo	AsgnDate	ProjNo	EmpNo	JobCode	AsgnHours
1001	12-Mar-16	15	103	503	3.5
1002	22-Mar-16	15	106	503	4.2
1003	22-Mar-16	22	102	504	2.0
1004	23-Mar-16	25	101	500	3.8
1005	12-Mar-16	18	104	502	5.6

- (i) Write **ONE (1)** SQL statement to retrieve the number of projects with the value within \$50,000.00 and \$100,000.00. (4 marks)
- (ii) Write **ONE (1)** SQL statement to display the employee's name, job description and day of the week on which the employee has been employed (label as Day). (4 marks)
- (iii) Write **ONE (1)** SQL statement to calculate the payment received for all employees according to their job assignment (label the column as Payment). Display your answer to show the employee's name, job description, assignment date and Payment. (5 marks)

[Total : 25 marks]

- Q2. (a) *Figure 2.1 shows the relations of Staff Branch database.*

Table Name: Staff				
StaffId	StaffTitle	StaffLName	StaffFName	BranchNo
A2021	Mr	Ogalio	Smith	1
A2120	Ms	Klington	Keith	3
A8922	Ms	Ohio	Sherry	5

Table Name: Branch			
BranchNo	BranchName	RegionCode	StaffId
1	Prime Corner	A1	A2021
2	Alley Glass Hope	A2	A2019
3	Hookies Junction	A1	A2120
4	Tuple Key Point	B1	A8922
5	Drive Angle	B2	A7722

Table Name: Region	
RegionCode	RegionName
A1	East
A2	West
B1	North
B2	South

Figure 2.1 Staff Branch Database

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- (i) For each table, identify the primary key and foreign key(s). If a table does not have a foreign key, write None. (6 marks)
 - (ii) Suggest the appropriate data type (Oracle Standard) for attributes StaffId and BranchNo from Staff and Branch tables respectively. (2 marks)
 - (iii) Name the parent and child tables in *Figure 2.1*. (3 marks)
- (b) The Entity Relationship Diagram (ERD) shows the Invoicing System of a company.

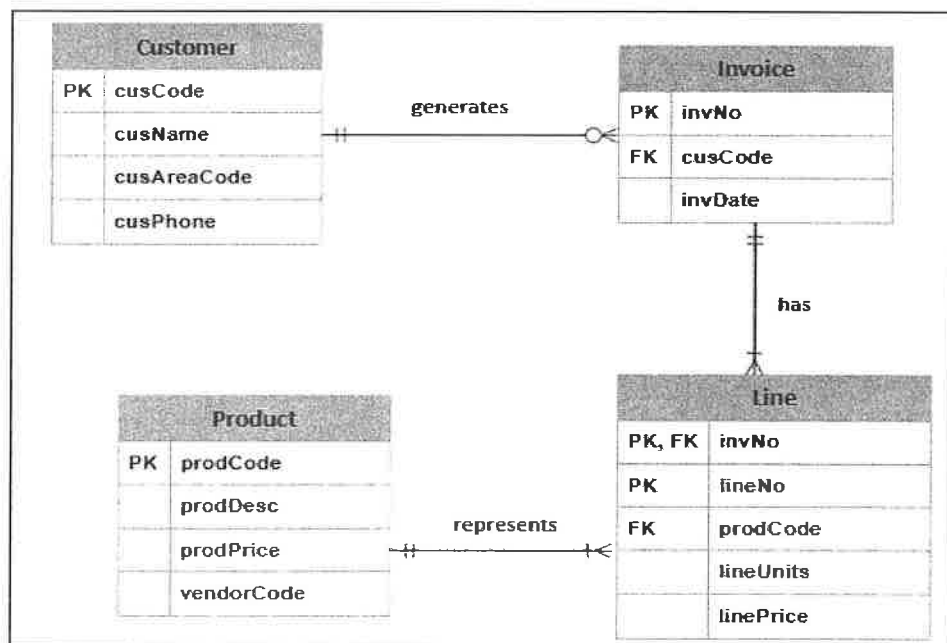


Figure 2.2 The partial entity relationship diagram (ERD) of Invoicing System

- (i) Identify **ONE (1)** derived attribute and describe how the value is derived. (4 marks)
- (ii) Specify **ONE (1)** weak entity from the relationship. Justify your answer. (3 marks)
- (iii) Identify **ONE (1)** example of multivalued attribute from the table. Justify your answer. (3 marks)
- (iv) Explain how entity integrity and referential integrity rules can be applied to these relations. (4 marks)

[Total : 25 marks]

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- Q3. (a) *Table 3.1* shows the sample data for books in the library. The data reflects that a book can be retrieved from several online sources.

Table 3.1 Sample Data of Books in Library

Attribute Name	Sample Value	Sample Value	Sample Value	Sample Value
bookID	121	332	434	516
bookName	Introduction to Programming Concepts	Psychology in Science Education	Discrete Mathematics and Technology	Advanced System and Engineering
shelfCode	ABB-12	BVV-98	ACC-43	ADD-33
shelfDept	Information Technology	Psychology	Science	Engineering
edition	2020	2019	2015	2022
sourceCode	IEEE, Springer, ADV	APPL, Springer	IEEE, DEV	TRAF, Springer, IEEE, DEV

- (i) Based on *Table 3.1*, identify whether the relation is in a First Normal Form (1NF). Explain your answer. (3 marks)
 - (ii) Describe **TWO (2)** rules to convert from First Normal Form (1NF) to Second Normal Form (2NF). (2 marks)
 - (iii) Based on the sample data shown in *Table 3.1*, draw the dependency diagram to show all dependencies in the relation. (5 marks)
 - (iv) Write the relational schema that meet 3NF requirements. Label the primary key and foreign key in each relation and create new entities and attributes as necessary. (6 marks)
- (b) *Figure 3.1* shows Employee's Club Enrollment Database that have been normalized to Third Normal Form (3NF) with three entities named as Enroll, Club and Employee.

Enroll				Club		
EmpID	ClubID	EnrolSDate	EnrolEDate	ClubID	CName	Role
MX20	B101	11/11/2021	11/11/2022	B101	Badminton	member
MZ18	T121	03/02/2019	03/02/2020	T121	Tennis	member
MZ18	B101	02/05/2023	02/05/2024	S144	Squash	member
PK17	S144	23/09/2017	23/09/2018	A878	Archery	member
MX20	A878	14/08/2021	14/08/2022			

Employee		
EmpID	EmpName	EmpDept
MX20	Lea Sofea	Finance
MZ18	Albert Wong	Production
PK17	Raymond Foo	Control Audit

Figure 3.1 Relation in Employee's Club Enrollment Database

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- (i) Draw the original table in First Normal Form (1NF). (8 marks)
 - (ii) Based on your answer in Q3(b)(i), identify the primary key of the relation. (1 mark)
- [Total : 25 marks]

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Section B [Total: 25 marks]. This section consists of **TWO (2) OPTIONAL** questions. Answer **ONE (1)** question only.

- Q4. (a) Several approaches can be considered when the tuple is going to be deleted from a relation with a child occurrence referencing to the parent occurrence. The approach will depend on the tables involved. Based on entity relationship diagram (ERD) in *Figure 4.1*, identify the parent and child relations, and indicate the appropriate actions that should be taken if the parent occurrence is deleted. Justify your answer. (5 marks)

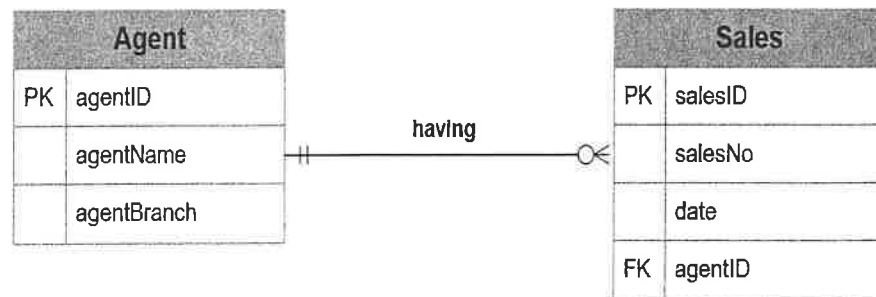


Figure 4.1 Entity relations of Agent and Sales

- (b) Define multivalued attributes and suggest **TWO (2)** possible actions that can be taken by database designer to handle the situation. (5 marks)
- (c) M&N Sdn Bhd is an agency that provides employment positions for potential employers. For that reason, the agency needs to keep information such as name, identity number, address, phone number and expertise of candidates who apply for job. Each candidate can list up to four types of expertise when applying the job. Besides that, the agency also needs to keep the employer-related records in the database. The request for hiring is submitted by the employer to the agency by stating the job position, salary offered and date of employment. The agency will then search for the number of qualified candidates to be selected for interview.
- (i) Draw **ONE (1)** ERD that includes all the entities, attributes, and relationships i.e., cardinality and optionality. For each entity, list at least **THREE (3)** relevant attributes. Use **CROW'S FOOT notation** or **UML notation** to illustrate your ERD. (10 marks)
- (ii) List the primary and foreign key(s) for each identified entity. Write **NONE** if no foreign key identified in the entity. (5 marks)

[Total : 25 marks]

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- Q5. (a) GH Automotive Inc. is producing specialty vehicles by contract and the company operates on several departments. The company is interested in creating a database to keep track of orders and to accelerate the process of delivering materials. Before a new vehicle is built, an order will be sent to the purchasing department to request for specific components. The order received by purchasing department may consist of several different items. An inventory is maintained so that the most frequently requested items are delivered almost immediately. When the order comes in, it is checked to determine whether the requested items are in the inventory. If an item is not in stock, the department will send the order to the supplier. Each item may require your company to liaise with several suppliers.
- (i) From the above case study, draw an entity relationship diagram (ERD) to include all the entities, attributes for each entity, relationships, primary key(s) and the multiplicity of the relationships, i.e., cardinality and optionality. Use **CROW'S FOOT notation** or **UML notation** to illustrate your ERD. (9 marks)
 - (ii) Based on the answer in Q5(a)(i), discuss **TWO (2)** types of existence dependence in all the relationships. (4 marks)
 - (iii) List **TWO (2)** examples of reports that can be generated from the database in the case study. (2 marks)
- (b) There are two types of connection traps that often occur due to misinterpretation of the meaning of a particular relationship between entities. Explain and draw a simple ERD to illustrate the following connection traps.
- (i) Chasm Trap. (5 marks)
 - (ii) Fan Trap. (5 marks)
- [Total : 25 marks]
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